Characterization of the PRs

omitted

2024-04-05

Data overview

```
glimpse(prs_characterization)
```

```
## Rows: 1,691
## Columns: 10
## $ pr_number          <int> 1302, 1304, 1313, 1322, 1329, 1332, 1343, 1346, 13…
## $ repo               <chr> "accumulo", "accumulo", "accumulo", "accumulo", "a…
## $ created_at         <dttm> 2019-07-31 19:34:39, 2019-08-01 18:52:50, 2019-08…
## $ merged_at          <dttm> 2019-08-02 15:59:29, 2019-08-01 20:56:42, 2019-08…
## $ additions          <int> 235, 89, 555, 112, 404, 1, 20, 14, 3, 9, 19, 27, 3…
## $ deletions          <int> 348, 91, 521, 109, 255, 1, 26, 30, 2, 8, 17, 31, 1…
## $ changed_files_count <int> 15, 5, 20, 5, 16, 1, 1, 1, 1, 1, 2, 1, 7, 7, 1, 11…
## $ commits_count      <int> 4, 2, 3, 1, 4, 1, 1, 2, 1, 1, 1, 3, 4, 5, 1, 3, 1,…
## $ code_churn         <int> 583, 180, 1076, 221, 659, 2, 46, 44, 5, 17, 36, 58…
## $ duration           <dbl> 1.85057870, 0.08601852, 6.99309028, 1.35975694, 9.…
```

PRs count

```
prs_characterization %>% count()
```

```
##      n
## 1 1691
```

PRs count by repo

```
prs_characterization %>%
    group_by(repo) %>%
    count()
```

```
## # A tibble: 12 × 2
## # Groups:   repo [12]
##    repo                      n
##    <chr>                 <int>
##  1 accumulo                793
##  2 cayenne                  54
##  3 commons-collections      22
##  4 commons-io               67
##  5 commons-lang            102
##  6 helix                   224
##  7 httpcomponents-client   117
##  8 maven-surefire           25
##  9 opennlp                  88
## 10 struts                  130
## 11 wicket                   40
## 12 zookeeper                29
```

Commits count

```
sum(prs_characterization$commits_count)
```

```
## [1] 3764
```

Commits count by repo

```
prs_characterization %>%
    group_by(repo) %>%
    summarise(sum(commits_count))
```

```
## # A tibble: 12 × 2
##    repo                  `sum(commits_count)`
##    <chr>                                <int>
##  1 accumulo                              1699
##  2 cayenne                                145
##  3 commons-collections                     46
##  4 commons-io                             174
##  5 commons-lang                           165
##  6 helix                                  691
##  7 httpcomponents-client                  202
##  8 maven-surefire                          37
##  9 opennlp                                110
## 10 struts                                 321
## 11 wicket                                  84
## 12 zookeeper                               90
```

General data statistics by repo

```
prs_characterization %>%
    group_by(repo) %>%
    summarise(
        min_date = min(merged_at),
        max_date = max(merged_at),
        age_in_days = difftime(min(merged_at), max(merged_at), units = "days")
    ) %>%
    mutate(
    age_in_days = abs(as.numeric(age_in_days)),
    age_in_years = abs(as.numeric(age_in_days))/365.25
    )
```

```
## # A tibble: 12 × 5
##    repo         min_date            max_date            age_in_days age_in_years
##    <chr>        <dttm>              <dttm>                    <dbl>        <dbl>
##  1 accumulo     2019-08-01 20:56:42 2024-02-28 21:32:19       1672.         4.58
##  2 cayenne      2021-12-07 10:13:35 2024-02-09 08:12:54        794.         2.17
##  3 commons-col… 2019-01-20 03:23:46 2023-07-27 19:16:34       1650.         4.52
##  4 commons-io   2016-12-02 02:08:46 2024-02-09 22:53:32       2626.         7.19
##  5 commons-lang 2016-11-22 21:25:59 2024-02-11 14:18:57       2637.         7.22
##  6 helix        2020-10-06 21:58:38 2024-02-22 18:13:28       1234.         3.38
##  7 httpcompone… 2018-11-14 19:18:37 2024-02-29 16:12:35       1933.         5.29
##  8 maven-suref… 2021-11-18 22:43:22 2023-08-23 16:41:09        643.         1.76
##  9 opennlp      2017-01-25 14:53:25 2024-01-13 13:53:26       2544.         6.96
## 10 struts       2018-11-20 11:45:50 2024-02-16 07:29:55       1914.         5.24
## 11 wicket       2018-08-01 03:29:48 2024-02-26 10:19:05       2035.         5.57
## 12 zookeeper    2022-01-26 12:23:10 2024-02-12 18:40:34        747.         2.05
```

```
prs_characterization %>%
    group_by(repo) %>%
    summarise(
        mean_commits = mean(commits_count),
        median_commits = median(commits_count),
        mean_changed_files = mean(changed_files_count),
        median_changed_files = median(changed_files_count)
    )
```

```
## # A tibble: 12 × 5
##    repo      mean_commits median_commits mean_changed_files median_changed_files
##    <chr>            <dbl>          <dbl>              <dbl>                <dbl>
##  1 accumulo          2.14            1                10.9                     3
##  2 cayenne           2.69            1                 8.07                    4
##  3 commons-…         2.09            1                13.8                     3
##  4 commons-…         2.60            2                 3.13                    2
##  5 commons-…         1.62            1                 3.30                    2
##  6 helix             3.08            2                 5.91                    3
##  7 httpcomp…         1.73            1                 6.14                    2
##  8 maven-su…         1.48            1                21.8                     3
##  9 opennlp           1.25            1                10.1                     4
## 10 struts            2.47            1.5              12.9                     4
## 11 wicket            2.1             1                 4.55                    2
## 12 zookeeper         3.10            2                 6.24                    2
```

```
prs_characterization %>%
    group_by(repo) %>%
    summarise(
        mean_code_churn = mean(code_churn),
        median_code_churn = median(code_churn),
        mean_duration = mean(duration)
    )
```

```
## # A tibble: 12 × 4
##    repo                  mean_code_churn median_code_churn mean_duration
##    <chr>                           <dbl>             <dbl>         <dbl>
##  1 accumulo                         382.              47            3.48
##  2 cayenne                          374.              68.5          5.78
##  3 commons-collections              132.              28            8.69
##  4 commons-io                        70.4             28           13.8
##  5 commons-lang                      62.8             10           17.4
##  6 helix                            227.              80            5.36
##  7 httpcomponents-client            188.              22            1.40
##  8 maven-surefire                   439.              22           20.9
##  9 opennlp                          690.              95            2.41
## 10 struts                          1088.             125            6.65
## 11 wicket                           101.              56            4.92
## 12 zookeeper                        265.              55           84.2
```

NCLOC and classes count by repo

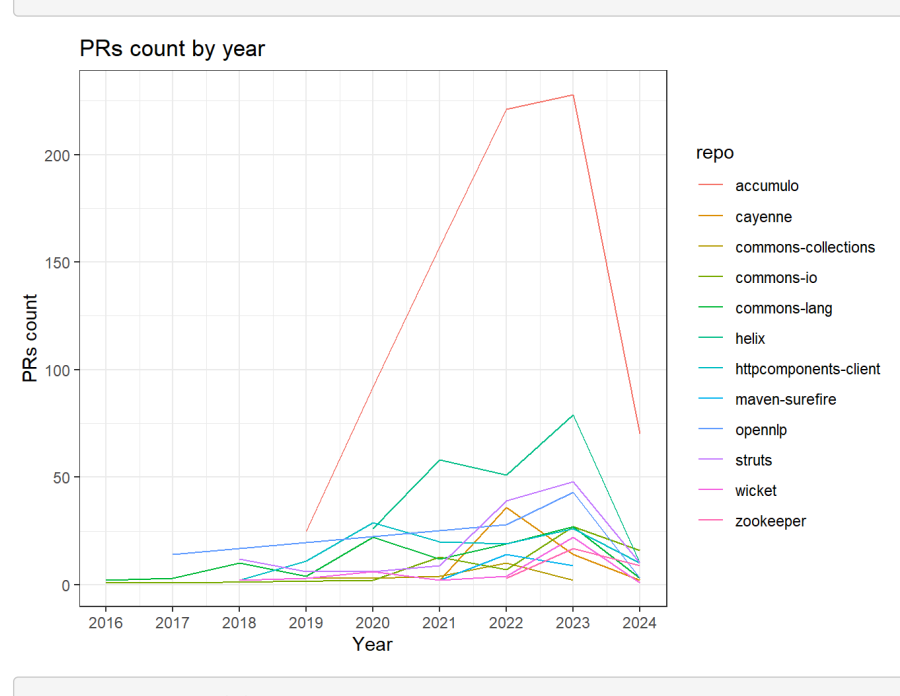
- All values are related to the last commit of the last PR that was successfully executed by SonarQube.

```
data.frame(repo = c("accumulo", "cayenne", "commons-collections","commons-io", "commons-lang", "helix", "httpcomponents-clie
nt", "maven-surefire", "opennlp", "struts", "wicket", "zookeeper"), NCLOC = c(440441, 318428, 67690, 30501, 95929, 189487, 7
6964, 110481, 155900, 234331, 251505, 131712), classes = c(5164, 4716, 839, 288, 918, 2106, 879, 3036, 2478, 3419, 5254, 154
2))
```

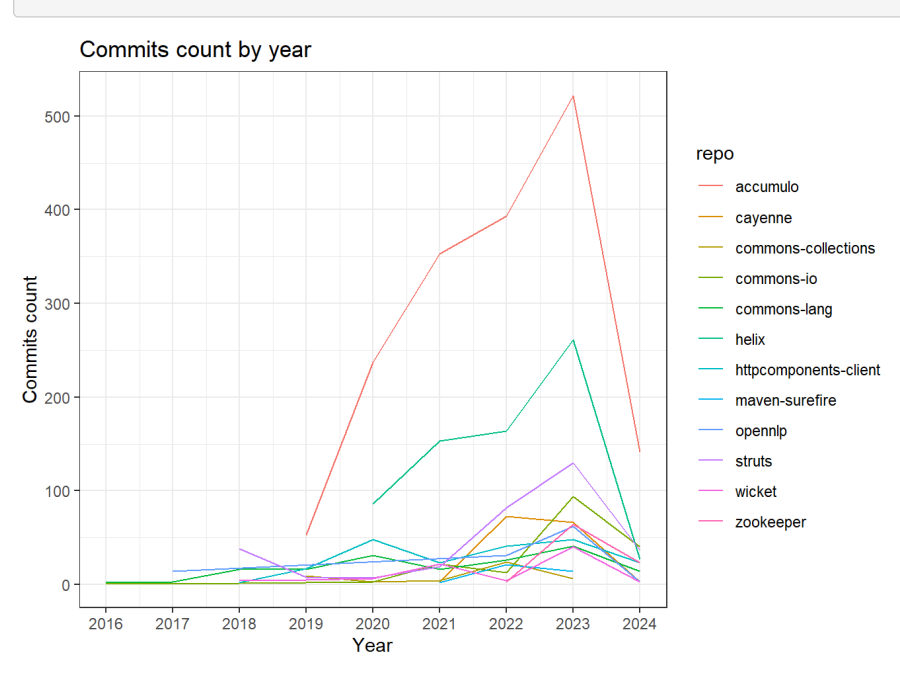
```
##                     repo  NCLOC classes
## 1               accumulo 440441    5164
## 2                cayenne 318428    4716
## 3    commons-collections  67690     839
## 4             commons-io  30501     288
## 5           commons-lang  95929     918
## 6                  helix 189487    2106
## 7  httpcomponents-client  76964     879
## 8         maven-surefire 110481    3036
## 9                opennlp 155900    2478
## 10                struts 234331    3419
## 11                wicket 251505    5254
## 12             zookeeper 131712    1542
```

PRs and commits count through the years

```
prs_characterization %>%
    mutate(year = lubridate::year(merged_at)) %>%
    group_by(year, repo) %>%
    summarise(PRs_quantity = n(), .groups = 'drop') %>%
    ggplot(aes(x = year, y = PRs_quantity, color=repo)) +
    geom_line() +
    labs(x = "Year", y = "PRs count") +
    ggtitle("PRs count by year") +
    scale_x_continuous(breaks = unique(lubridate::year(prs_characterization$merged_at)))
```



```
prs_characterization %>%
    mutate(year = lubridate::year(merged_at)) %>%
    group_by(year, repo) %>%
    summarise(commits_quantity = sum(commits_count), .groups = 'drop') %>%
    ggplot(aes(x = year, y = commits_quantity, color=repo)) +
    geom_line() +
    labs(x = "Year", y = "Commits count") +
    ggtitle("Commits count by year") +
    scale_x_continuous(breaks = unique(lubridate::year(prs_characterization$merged_at)))
```



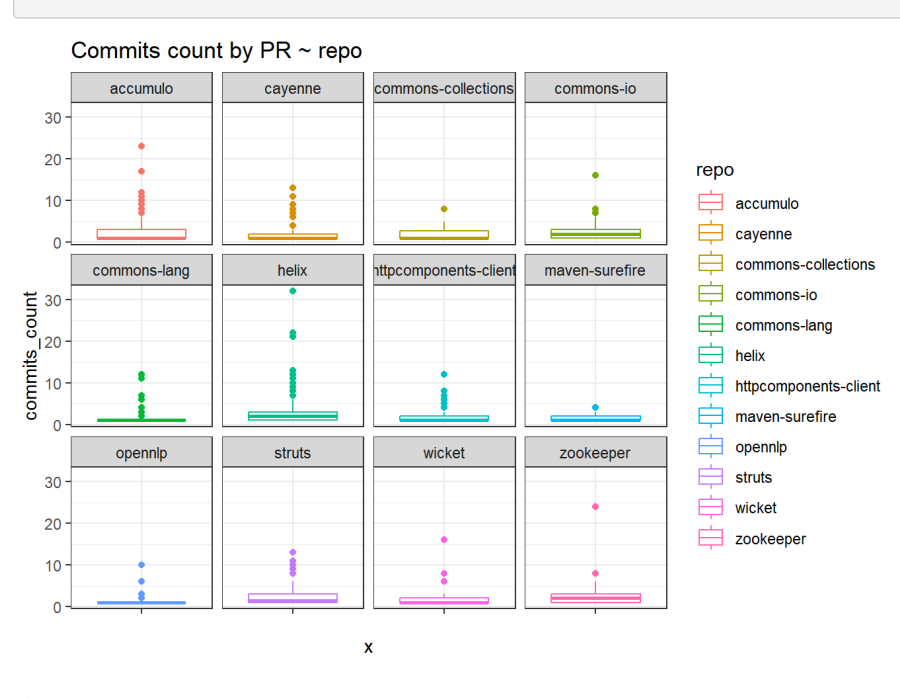
Distribution of features

Commits count by PR

```
tapply(prs_characterization$commits_count, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   2.142   3.000  23.000 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   2.685   2.000  13.000 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   2.091   2.750   8.000 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   2.597   3.000  16.000 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   1.618   1.000  12.000 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   3.085   3.000  32.000 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.000   1.726   2.000  12.000 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.00    1.00    1.48    2.00    4.00 
## 
## $opennlp
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.00    1.00    1.25    1.00   10.00 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   1.500   2.469   3.000  13.000 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0     1.0     1.0     2.1     2.0    16.0 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   3.103   3.000  24.000
```

```
prs_characterization %>%
    ggplot(aes(x = "", y = commits_count, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Commits count by PR ~ repo")
```

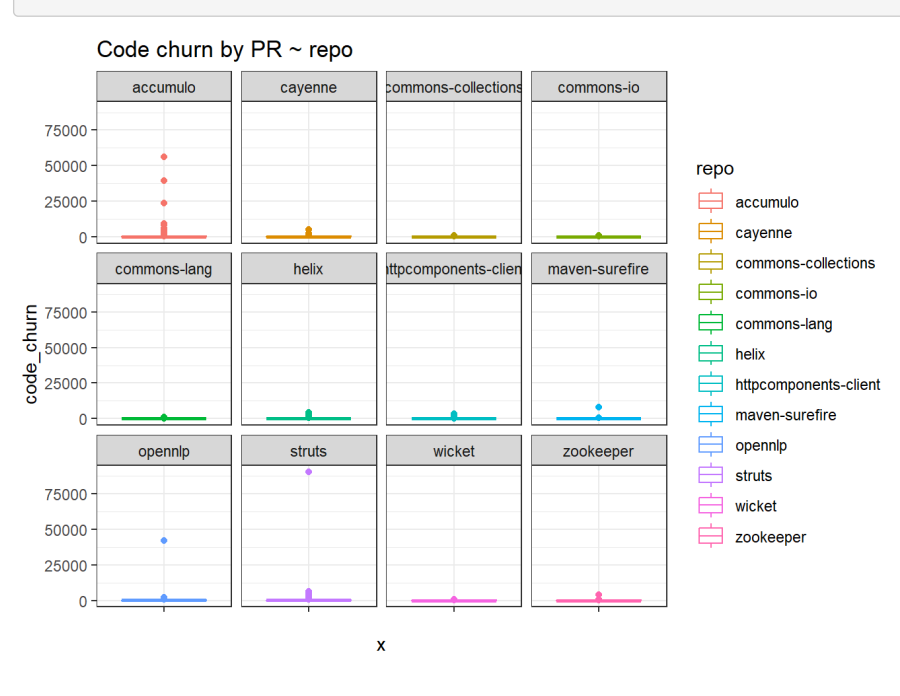


Code churn by PR

```
tapply(prs_characterization$code_churn, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0    12.0    47.0   381.8   163.0 55972.0 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0    26.5    68.5   374.5   376.8  4793.0 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0    10.0    28.0   132.2    75.0  1035.0 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    6.50   28.00   70.45   61.00  631.00 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    4.00   10.00   62.76   29.75 1046.00 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0    25.5    80.0   227.2   203.2  4343.0 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0     4.0    22.0   187.8    97.0  3259.0 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0     8.0    22.0   438.9   168.0  8229.0 
## 
## $opennlp
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
##     2.00    17.75    95.00   690.10   273.00 41938.00 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0    39.0   125.0  1088.2   297.2 90610.0 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     3.0    16.5    56.0   100.8   117.2   849.0 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     2.0     6.0    55.0   265.3   135.0  4005.0
```

```
prs_characterization %>%
    ggplot(aes(x = "", y = code_churn, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Code churn by PR ~ repo")
```

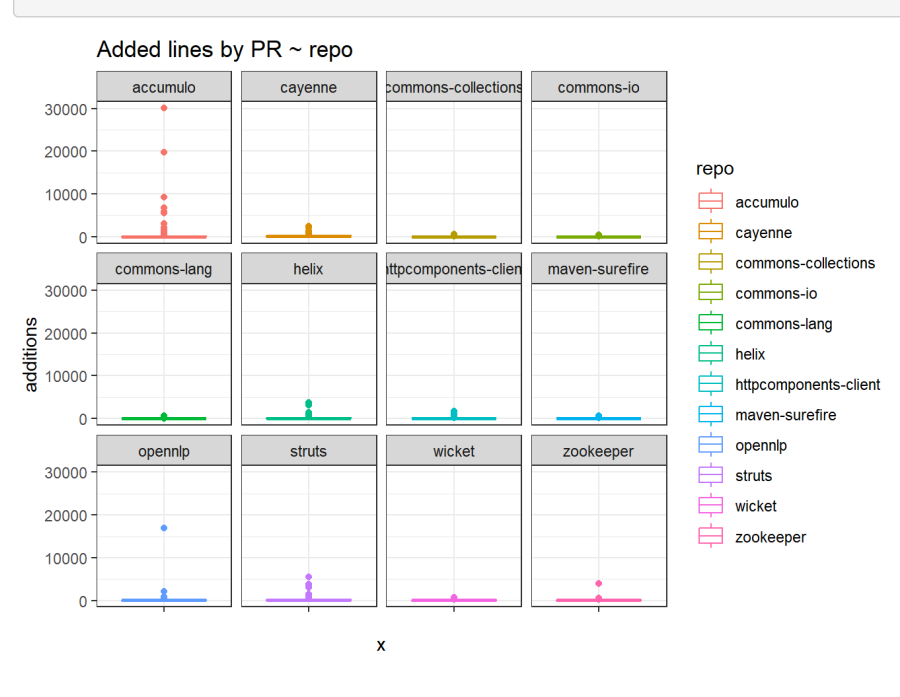


Added lines

```
tapply(prs_characterization$additions, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     6.0    24.0   195.1    92.0 30103.0 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0    17.5    53.0   263.1   274.2  2447.0 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    5.50   25.50   57.59   40.00  536.00 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     4.0    20.0    38.6    43.0   399.0 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    2.00    6.00   35.65   18.50  591.00 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00   16.75   53.00  177.06  150.00 3718.00 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     2.0    13.0   109.7    80.0  1748.0 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    3.00    9.00   70.44   47.00  579.00 
## 
## $opennlp
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
##     0.00     8.75    58.50   337.08   158.50 16944.00 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00   12.75   71.00  269.46  200.25 5565.00 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    7.75   37.00   72.60   75.00  650.00 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     1.0     3.0    49.0   225.7   103.0  4004.0
```

```
prs_characterization %>%
    ggplot(aes(x = "", y = additions, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Added lines by PR ~ repo")
```

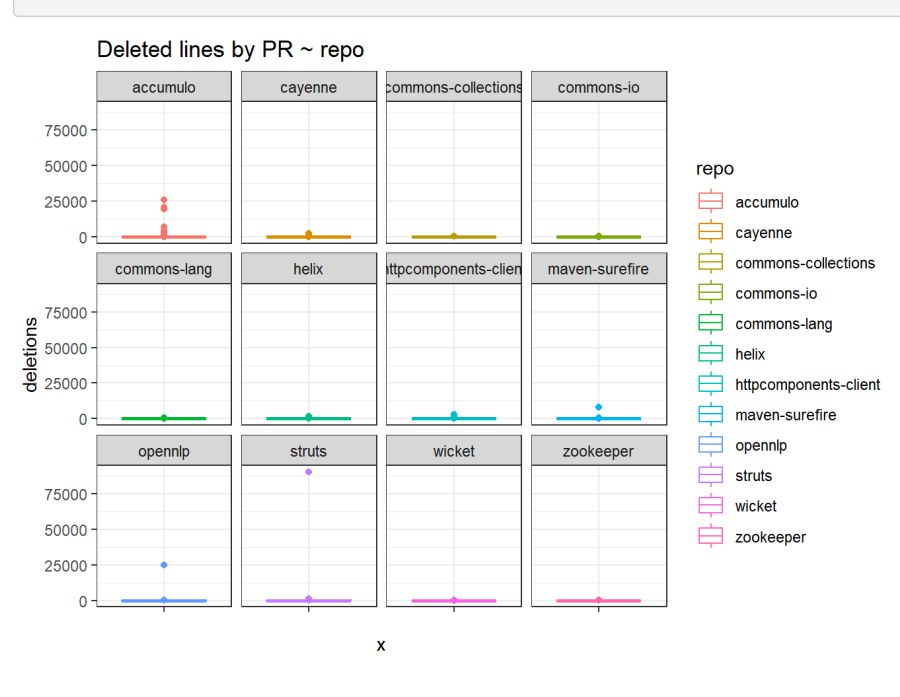


Deleted lines

```
tapply(prs_characterization$deletions, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     3.0    14.0   186.7    54.0 25869.0 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     2.0    13.0   111.4    39.0  2346.0 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    1.00    9.00   74.64   37.00  547.00 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    1.00    6.00   31.85   20.00  423.00 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    1.00    3.00   27.12   10.00  527.00 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    3.00   13.00   50.12   37.00 1506.00 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    1.00    3.00   78.09   27.00 2860.00 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     3.0     9.0   368.4    20.0  8228.0 
## 
## $opennlp
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##       0       2      20     353     107   24994 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##     0.0     6.0    28.5   818.7   115.8 90609.0 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    0.00    3.75   11.00   28.25   25.25  199.00 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.00    4.00   39.59   50.00  307.00
```

```
prs_characterization %>%
    ggplot(aes(x = "", y = deletions, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Deleted lines by PR ~ repo")
```

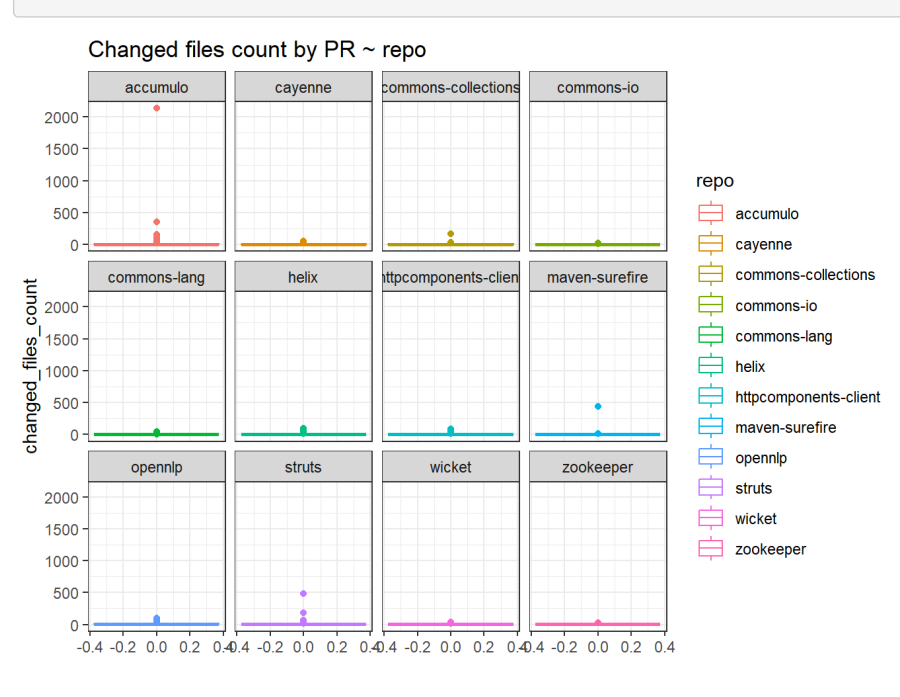


Changed files count

```
tapply(prs_characterization$changed_files_count, prs_characterization$repo, summary)
```

```
## $accumulo
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.00    3.00   10.89    7.00 2132.00 
## 
## $cayenne
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   2.000   4.000   8.074   9.500  57.000 
## 
## $`commons-collections`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.25    3.00   13.82   11.25  165.00 
## 
## $`commons-io`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   3.134   3.000  19.000 
## 
## $`commons-lang`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   3.304   2.000  42.000 
## 
## $helix
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   2.000   3.000   5.906   6.000  97.000 
## 
## $`httpcomponents-client`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   6.137   4.000  85.000 
## 
## $`maven-surefire`
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    3.00   21.84    6.00  434.00 
## 
## $opennlp
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    4.00   10.09   11.25  103.00 
## 
## $struts
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    2.00    4.00   12.87    8.75  484.00 
## 
## $wicket
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    1.00    1.00    2.00    4.55    4.25   32.00 
## 
## $zookeeper
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##   1.000   1.000   2.000   6.241  10.000  29.000
```

```
prs_characterization %>%
    ggplot(aes(y = changed_files_count, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "Changed files count by PR ~ repo")
```

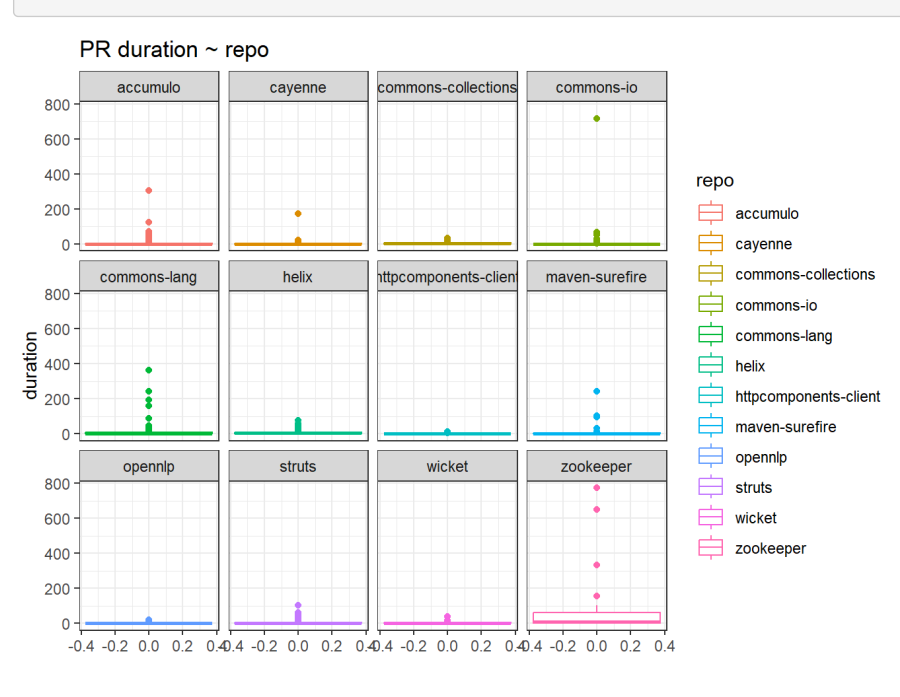


PR duration

```
tapply(prs_characterization$duration, prs_characterization$repo, summary)
```

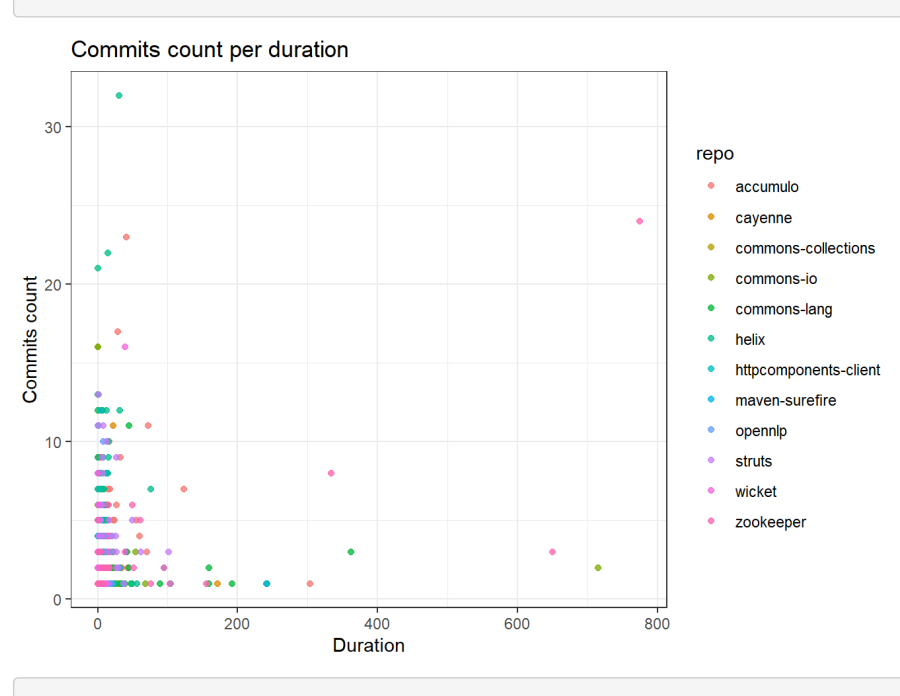
```
## $accumulo
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.00316   0.14716   0.82903   3.48137   2.94471 303.90714 
## 
## $cayenne
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
## 8.924e-03 3.408e-01 9.842e-01 5.777e+00 3.406e+00 1.717e+02 
## 
## $`commons-collections`
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##  0.004421  0.282948  2.981742  8.692700  7.277911 34.237477 
## 
## $`commons-io`
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.03005   0.14458   0.33830  13.78381   1.00329 715.08228 
## 
## $`commons-lang`
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.00301   0.12904   0.77149  17.35842   7.03443 362.21278 
## 
## $helix
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##  0.000174  0.834734  2.838843  5.356707  6.764641 76.235810 
## 
## $`httpcomponents-client`
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##  0.002651  0.175116  0.582789  1.396536  1.910116 12.367778 
## 
## $`maven-surefire`
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
##   0.1133   0.5058   1.2199  20.8596   4.4510 240.9541 
## 
## $opennlp
##     Min.  1st Qu.   Median     Mean  3rd Qu.     Max. 
##  0.00419  0.43870  1.13698  2.41061  2.75728 20.87878 
## 
## $struts
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.01024   0.70001   2.67966   6.64991   6.98276 101.57461 
## 
## $wicket
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##  0.007164  0.444751  1.581991  4.918542  5.365738 39.144803 
## 
## $zookeeper
##      Min.   1st Qu.    Median      Mean   3rd Qu.      Max. 
##   0.07884   0.84983   8.83714  84.16486  60.90042 774.56779
```

```
prs_characterization %>%
    ggplot(aes(y = duration, color=repo)) +
      geom_boxplot() +
      facet_wrap(~repo) +
      labs(title = "PR duration ~ repo")
```

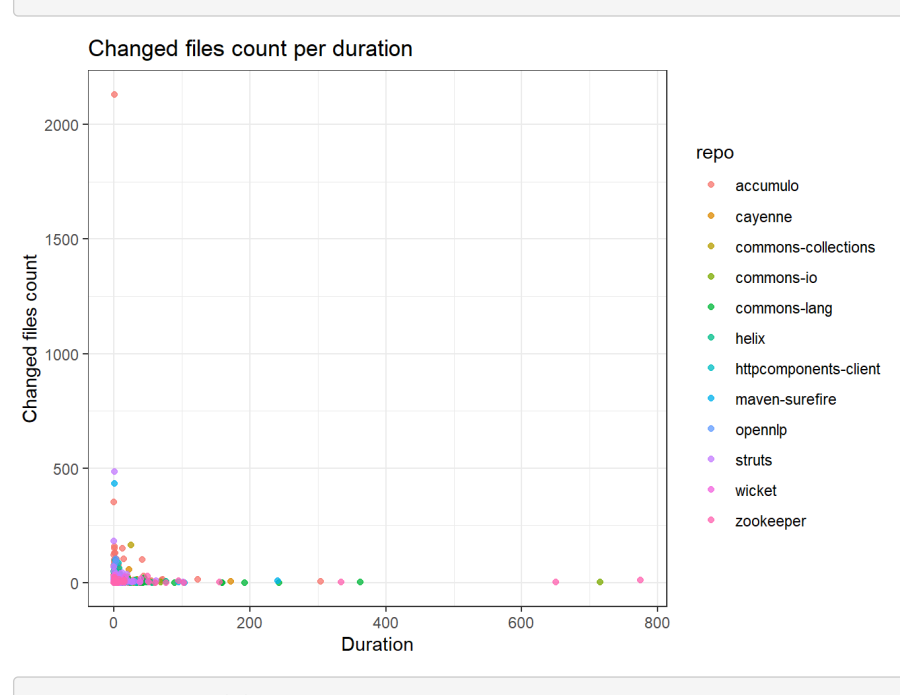


Scatters

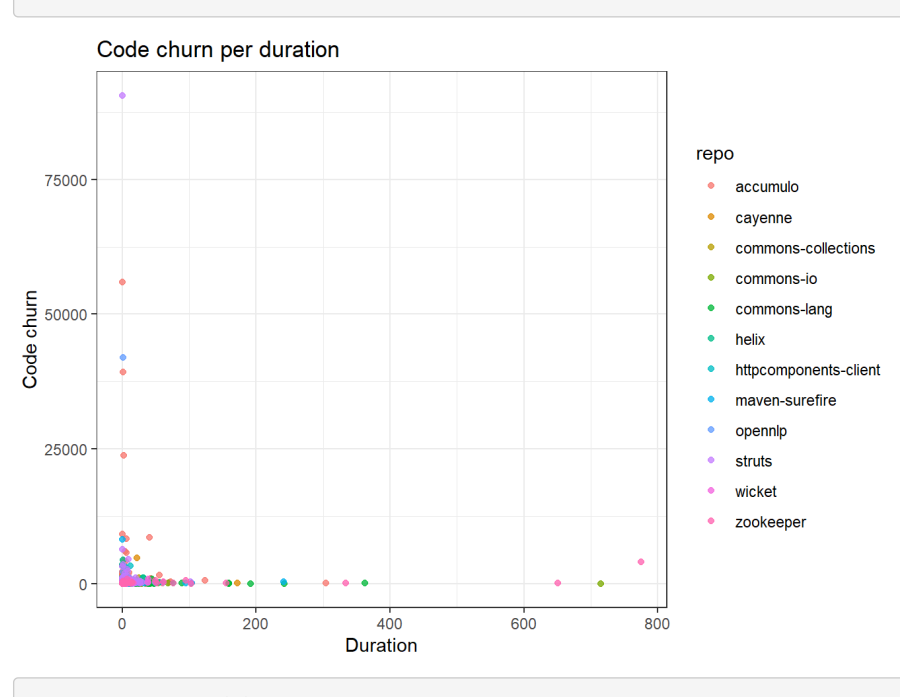
```
prs_characterization %>%
    ggplot(aes(x = duration, y = commits_count, color=repo)) +
    geom_point(alpha=0.75) +
    labs(x = "Duration", y = "Commits count") +
    ggtitle("Commits count per duration")
```



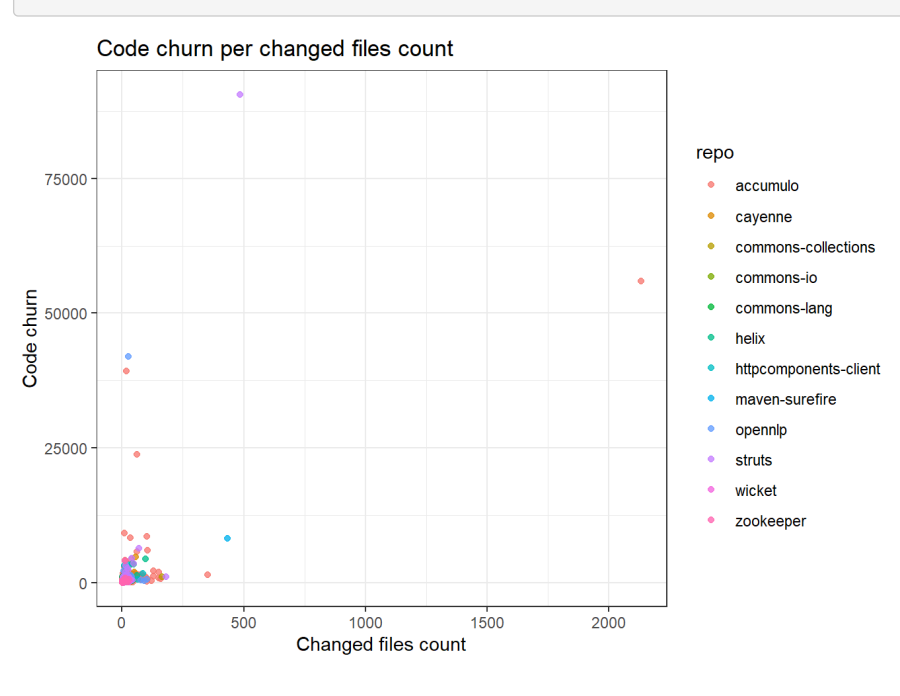
```
prs_characterization %>%
    ggplot(aes(x = duration, y = changed_files_count, color=repo)) +
    geom_point(alpha=0.75) +
    labs(x = "Duration", y = "Changed files count") +
    ggtitle("Changed files count per duration")
```



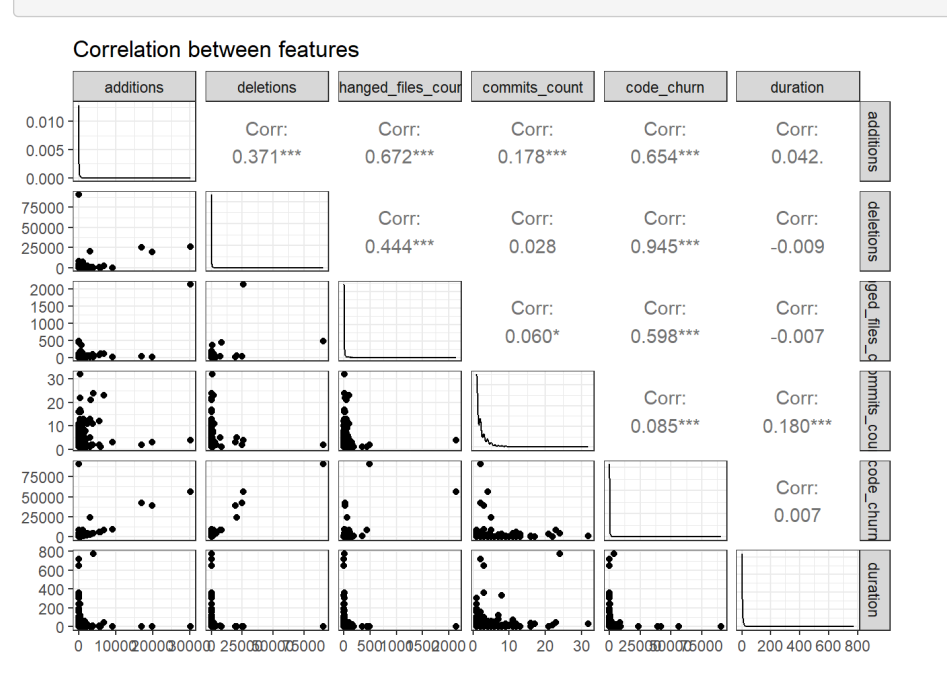
```
prs_characterization %>%
    ggplot(aes(x = duration, y = code_churn, color=repo)) +
    geom_point(alpha=0.75) +
    labs(x = "Duration", y = "Code churn") +
    ggtitle("Code churn per duration")
```



```
prs_characterization %>%
    ggplot(aes(x = changed_files_count, y = code_churn, color=repo)) +
    geom_point(alpha=0.75) +
    labs(x = "Changed files count", y = "Code churn") +
    ggtitle("Code churn per changed files count")
```



Correlations

```
prs_characterization %>%
    select(additions, deletions, changed_files_count, commits_count, code_churn, duration) %>%
    ggpairs(title="Correlation between features")
```